

Experiment Results Summary (Exp 09-13)

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Summary Table:

Exp	Model	Accuracy	Status
09	RGB_EfficientNetB0-Laplacian	0.827	FAILED (val_acc=0.50)
10	RGB-EfficientNetB0 (Roselab + Android-captured)	0.922	GOOD
11	RGB-EnetB0 (NTU-Android x8) + Testing iPhone	0.922	CONSISTENT
12	Robust_3_devices_EfficientNetB0	0.948	BEST
13	MobileNetV3_3_Mobile_devices	0.890	GOOD

Key Observations:

- 1. Experiment 09 failed with validation accuracy 50% (random guessing).
- 2. Experiments 10 and 11 report identical metrics (0.922 accuracy, 566 samples).
- 3. Experiment 12 is the best performing model (0.948 accuracy on 135 samples).
- 4. Experiment 13 is an efficient alternative with 0.890 accuracy using MobileNetV3.
- 5. Successful experiments (10-13) have balanced precision/recall scores.
- 6. Dataset sizes vary: 81 (Exp 09), 566 (Exp 10-11), 135 (Exp 12-13).

Experiment 09 - RGB_EfficientNetB0-Laplacian

Status: FAILED (val_acc=0.50)

Reported Accuracy: 0.8272

Class	Precision	Recall	F1-score	Support
Original	0.8649	0.7805	0.8205	41
Recaptured	0.7955	0.8750	0.8333	40

Macro Avg (P/R/F1): 0.8302 / 0.8277 / 0.8269

Weighted Avg (P/R/F1): 0.8306 / 0.8272 / 0.8268

JSON Data:

```
{
  "val_accuracy": 0.82,
  "original_correct": 32,
  "recaptured_correct": 35,
  "total": 81
}
```

Experiment 10 - RGB-EfficientNetB0 (Roselab + Android-captured)

Status: GOOD

Reported Accuracy: 0.9223

Class	Precision	Recall	F1-score	Support
Original	0.9300	0.9100	0.9200	284

Recaptured	0.9100	0.9300	0.9200	282
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Macro Avg (P/R/F1): 0.9200 / 0.9200 / 0.9200

Weighted Avg (P/R/F1): 0.9200 / 0.9200 / 0.9200

JSON Data:

```
{
  "val_accuracy": 0.9223,
  "f1_recaptured": 0.9228,
  "original_correct": 259,
  "recaptured_correct": 263,
  "total_samples": 566
}
```

Experiment 11 - RGB-EnetB0 (NTU-Android x8) + Testing iPhone

Status: CONSISTENT

Reported Accuracy: 0.9223

Class	Precision	Recall	F1-score	Support
Original	0.9300	0.9100	0.9200	284
Recaptured	0.9100	0.9300	0.9200	282

Macro Avg (P/R/F1): 0.9200 / 0.9200 / 0.9200

Weighted Avg (P/R/F1): 0.9200 / 0.9200 / 0.9200

JSON Data:

```
{
  "val_accuracy": 0.9223,
  "f1_recaptured": 0.9228,
  "original_correct": 259,
  "recaptured_correct": 263,
  "total_samples": 566
}
```

Experiment 12 - Robust_3_devices_EfficientNetB0

Status: BEST

Reported Accuracy: 0.9481

Class	Precision	Recall	F1-score	Support
Original	0.9300	0.9700	0.9500	68
Recaptured	0.9700	0.9300	0.9500	67

Macro Avg (P/R/F1): 0.9500 / 0.9500 / 0.9500

Weighted Avg (P/R/F1): 0.9500 / 0.9500 / 0.9500

JSON Data:

```
{
  "accuracy": 0.9481,
  "f1_original": 0.9496,
  "f1_recaptured": 0.9466
}
```

Experiment 13 - MobileNetV3_3_Mobile_devices

Status: GOOD
Reported Accuracy: 0.8900

Class	Precision	Recall	F1-score	Support
Original	0.9000	0.8800	0.8900	68
Recaptured	0.8800	0.9000	0.8900	67

Macro Avg (P/R/F1): 0.8900 / 0.8900 / 0.8900
Weighted Avg (P/R/F1): 0.8900 / 0.8900 / 0.8900

JSON Data:
null

Final Summary and Recommendations

Summary:

- Experiments 10 and 11 show identical, stable performance across datasets (0.922 accuracy).
- Experiment 12 achieves the best performance (0.948 accuracy) and is recommended for production.
- Experiment 09 shows failure on validation (50% val accuracy). Investigate dataset or preprocessing differences.
- Experiment 13 (MobileNetV3) is a good lightweight alternative (0.890 accuracy).

Recommendations:

1. Prioritize model from Experiment 12 for deployment (Robust_3_devices_EfficientNetB0).
2. Use Experiment 13 when compute or memory is constrained on device.
3. Re-evaluate Laplacian preprocessing used in Experiment 09 that led to validation collapse.
4. Maintain an identical test set across experiments when possible to reduce comparison variance.
5. Collect more diverse recapture samples to reduce remaining error margins.